Large Sample (*Z-Tools*) Tests

If we have an estimate such that $\hat{\theta} \approx Normal(\mu_{\hat{\theta}} = \theta, \sigma^2_{\hat{\theta}}),$ we have the following pivotal quantity,

$$Z = \frac{\hat{\theta} - \mu_{\hat{\theta}}}{\sigma_{\hat{\theta}}} = \frac{\hat{\theta} - \theta}{\sigma_{\hat{\theta}}} \approx Normal(0,1).$$

Test Set - up :

$$H_0 : \theta = \theta_0 \quad \text{vs.} \quad H_A : \begin{cases} i) \quad \theta > \theta_0 & \text{(called a right-tailed test)} \\ ii) \quad \theta < \theta_0 & \text{(called a left-tailed test)} \\ iii) \quad \theta \neq \theta_0 & \text{(called a two-tailed test)} \end{cases}$$

Note: The alternative (or research) hypothesis is determined based on the research question.

Large-Sample (*Z-Tools*) Tests: Applications

Parameter: θ	Sample Size(s)	Estimator: $\hat{\theta}$	$\sigma^2_{\hat{\theta}}$	$\hat{\sigma}^2_{\hat{\theta}}$
μ	n	$\bar{Y}$	$\frac{\sigma^2}{n}$	$\frac{S^2}{n}$
$\mu_1 - \mu_2$	n_1 and n_2	$\bar{Y}_1 - \bar{Y}_2$	$\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}$	$\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}$
p	n	$\hat{p}$	$\frac{p(1-p)}{n}$	$\frac{\hat{p}(1-\hat{p})}{n}$
$p_1 - p_2$	n_1 and n_2	$\hat{p}_1 - \hat{p}_2$	$\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}$	$\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}$

Assumptions:

- We have large sample sizes.
  However, in the case of means, these intervals apply if we are sampling from Normal populations with known variances for any sample size(s).

- In the multiple sample applications, the samples are independent.

Let's explore the right-tailed *Z*-test:

Test Set - up :

$H_0 : \theta = \theta_0$ vs. $H_A : \theta > \theta_0$

Consider the rejection region $RR = \{\hat{\theta} > k\}$.

1. Given α, the significance level of the test, what is k?

2. What is the p - $value$ of the test?

*Repeat for the left-tailed and two-tailed tests.

Summary of *Z*-tests:

- Parameter: θ
- Consider testing $H_0 : \theta = \theta_0$ vs. $H_A : \begin{cases} i) & \theta > \theta_0 \\ ii) & \theta < \theta_0 \\ iii) & \theta \neq \theta_0 \end{cases}$
- Sampling Distribution of $\hat{\theta}$: If $\hat{\theta} \approx Normal(\mu_{\hat{\theta}} = \theta, \sigma^2_{\hat{\theta}})$, then

$$\frac{\hat{\theta} - \theta}{\sigma_{\hat{\theta}}} \approx Normal(0,1).$$

- Test-Statistic:

$$TS = z_0^* = \frac{\hat{\theta} - \theta_0}{\hat{\sigma}_{\hat{\theta}}}.$$

- $RR_\alpha = \begin{cases} i) & \{z_0^* > z_\alpha^*\} \\ ii) & \{z_0^* < -z_\alpha^*\} \\ iii) & \{|z_0^*| > z_{\alpha/2}^*\} \end{cases}$, $p-value = \begin{cases} i) & P(Z > z_0^*) \\ ii) & P(Z < z_0^*) \\ iii) & 2P(Z > z_0^*) \end{cases}$

Example 1d: (Large Sample Z-test for a population proportion)

An experimenter has prepared a drug dosage level that she claims will induce sleep for 80% of insomniacs. After examining the dosage, a consumer group feels the claim regarding the effectiveness of the drug has been inflated. In an attempt to disprove her claim, the consumer group administers the drug to 50 insomniacs. They observe that the drug induced sleep for 32 of these individuals.

The test set up is:

$$H_0 : p = 0.8 \quad vs. \quad H_a : p < 0.8$$

Use the large sample approximation to carry out the test.

Type II Error and Power of the Test

Recall: Decision Errors:

1. Type I Error: Rejecting a true H_0.

   Define: • $\alpha = P(\text{Type I Error}) = P(\text{Reject } H_0 \mid H_0 \text{ is true})$.

2. Type II Error: Failing to reject a false H_0.

   Define: $\beta = P(\text{Type II Error}) = P(\text{Fail to reject } H_0 \mid H_0 \text{ is false})$.

Now define: **Power of the Test**

Consider some value of θ under H_a, θ_a. Then the power of the test is defined as

$$\Rightarrow power(\theta_a) = P(\text{Reject } H_0 \mid \underbrace{\theta = \theta_a}_{H_0 \text{ is false}}) = P(\hat{\theta} \in RR \mid \underbrace{\theta = \theta_a}_{H_0 \text{ is false}}) = 1 - \beta_{\theta_a}.$$

Power of the right-tailed *Z*-test:

Test Set - up :

$H_0 : \theta = \theta_0$ vs. $H_a : \theta > \theta_0$

• For a chosen α, $RR_\alpha = \{\hat{\theta} > \theta_0 + z_\alpha^* \sigma_{\hat{\theta}}\} = \{z_0^* > z_\alpha^*\}$.

$\Rightarrow$ Find a probability expression for $power(\theta_a)$.

*Repeat for the left-tailed and the two-tailed tests.

Example 4: The amount filled (in ounces) by a certain bottling machine is a random variable following approximately a Normal distribution with unknown mean and standard deviation equal to 2 ounces. Suppose a sample of size n=9 resulted in a sample mean of 14.5 ounces. Is this an indication that the bottling machine is filling bottles with mean fill less than 16 ounces?

H_0 : (no) $\mu_Y = 16$
H_a : (yes) $\mu_Y < 16$

- Find $power(\mu_a)$. Explore this for different values of α and μ_a.

- Find the value of k so that $\alpha = \beta$.

- **Sample Size Calculation :** Suppose we want to carry out the test with chosen levels of α and β_{μ_a}. How many observations do we need?